

MD ABU YOUSUF PRODHAN

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PHYSICAL OCEANOGRAPHY • LAGRANGIAN TRANSPORT • COASTAL CIRCULATION • NUMERICAL MODELLING •
FIELD-MODEL INTEGRATION

RESEARCH PROFILE

Physical oceanographer with coursework in partial differential equations, vector calculus, and linear algebra. Experienced in Lagrangian particle-tracking and coastal trajectory simulation, seeking to combine numerical modelling with field-based drifter validation to understand transport, dispersion, and residence-time processes in fjord and estuarine systems. Incoming NF-POGO Scholar at Dalhousie University.

EDUCATION

Post-Graduation in Observational Oceanography

Ocean Frontier Institute, Dalhousie University

Halifax, Canada

Oct 2026 – Jul 2027

NF-POGO Centre of Excellence Scholar. Fully funded 10-month postgraduate training program (Nippon Foundation); selected as 1 of 10 international scholars in the 2026–27 cohort.

M.Sc. in Oceanography

Shahjalal University of Science and Technology (SUST)

Sylhet, Bangladesh

Jul 2025 – Dec 2026

CGPA: **3.99/4.00** Supervisor: Dr. Subrata Sarker

Thesis: “Identifying Ecological Drivers of Phytoplankton Functional Diversity Using Field Observations and Machine Learning in the Coastal Ecosystem of Bangladesh”

B.Sc. (Honors) in Oceanography

Shahjalal University of Science and Technology (SUST)

Sylhet, Bangladesh

Feb 2020 – May 2025

CGPA: **3.81/4.00** Supervisor: Dr. Subrata Sarker

Coursework: Calculus and Vector Analysis; Matrix and Differential Equations (including partial differential equations); Linear Algebra

Term Paper: “Modelling the Fate and Trajectories of Hypothetical Oil Spills Near Chattogram and Mongla Ports, Bangladesh” [\[Link\]](#)

RESEARCH EXPERIENCE

M.S. Thesis

Jul 2025 – Dec 2026

Department of Oceanography, SUST, Bangladesh

Quantified phytoplankton functional diversity (FRic, FEve, FDiv, FDis, Rao’s Q) from field observations in Bangladesh coastal waters.

Built machine-learning frameworks (Random Forest, XGBoost) to identify ecological drivers of community assembly and trait-environment relationships.

Integrated in-situ pigment data with satellite-derived covariates to assess seasonal controls on phytoplankton functional structure.

International Research Intern

Feb 2026

Centre for Policy Research, Universiti Sains Malaysia, Penang, Malaysia

[\[Link\]](#)

Contributed to collaborative marine governance and blue-carbon policy research between USM and SUST.

Conducted coastal field activities including seagrass habitat observation and environmental data collection.

Independent Research Projects

2023 – 2025

Department of Oceanography, SUST, Bangladesh

Oil-Spill Trajectory Modelling. Simulated hypothetical spills in the Bay of Bengal using Open-Drift/OpenOil forced with CMEMS wind, wave, and current reanalysis; processed multi-dimensional NetCDF outputs in Python. Identified that trajectory accuracy was constrained by the spatial resolution of available reanalysis forcing data, motivating interest in higher-resolution regional ocean modelling and direct drifter-based field validation of Lagrangian trajectories. [\[GitHub\]](#)

HAB Early-Warning System. Developed a satellite-based MODIS-Aqua pipeline (2010–2024) for harmful algal bloom risk forecasting using Random Forest and XGBoost; produced seasonal risk maps for the northern Bay of Bengal. [\[GitHub\]](#)

Phytoplankton & POC Dynamics. Analyzed MODIS-Aqua chlorophyll-*a* and POC concentrations using SNAP and R; examined seasonal productivity patterns and blue-green band-ratio algorithms across the Bay of Bengal. [\[Link\]](#)

| PUBLICATIONS

Prodhan et al. (2025) Modelling the fate and trajectories of hypothetical oil spills near Chattogram and Mongla Port, Bangladesh. *International Conference on Marine Spatial Planning for Advancing Sustainable Blue Growth*, 2025. [\[Link\]](#) [\[PDF\]](#)

Prodhan & Ahmad (2024). Satellite-based assessment of particulate organic carbon (POC) concentrations in the Bay of Bengal: Spatial and temporal dynamics. *Manuscript under peer review*. [\[Preprint\]](#)

| PRESENTATIONS

Prodhan et al. (2025) Modelling the fate and trajectories of hypothetical oil spills near Chattogram and Mongla Port, Bangladesh. *International Conference on Marine Spatial Planning for Advancing Sustainable Blue Growth*, 2025. Oral presentation.

| FELLOWSHIPS & AWARDS

NF-POGO Centre of Excellence Scholar (2026–27), Ocean Frontier Institute, Dalhousie University. Competitive international selection (10 scholars worldwide).

Trevor Platt Science Foundation Mentee (2025), mentored by Dr. Sebastian Krieger in physical oceanography and quantitative modelling. [\[Link\]](#)

Champion, NOAMI Ocean Olympiad 2.0 (2025), National Oceanographic and Maritime Institute, Bangladesh. 1st place among 200+ participants. [\[Link\]](#)

Millennium Fellow, Class of 2024, UN Academic Impact & Millennium Campus Network. Selected from 52,000+ applicants. [\[Link\]](#)

Aspire Leadership Program, Aspire Institute & Harvard Business School (2025). 40 hours of training in critical thinking and social impact. [\[Link\]](#)

| TECHNICAL SKILLS

Mathematical Methods	Partial Differential Equations, Vector Calculus, Linear Algebra, Fourier Analysis, Time-Series Analysis
Programming	Python, R, NumPy, Pandas, Matplotlib, Scikit-learn, Google Colab
Statistical Modelling	GAMs, Random Forest, XGBoost, SHAP interpretation, time-series analysis
Ocean Modelling	OpenDrift/OpenOil, Delft3D FM, Lagrangian particle tracking, NetCDF, CMEMS products
Remote Sensing	MODIS, Landsat, Sentinel, SNAP, ArcGIS, Google Earth Engine; ocean-color products (Chl- <i>a</i> , POC, SST)
Biogeochemistry	Carbon and nutrient cycling, phytoplankton trait-based ecology, functional diversity indices (FRic, FEve, FDiv, FDis, Rao's Q)

| LANGUAGES

English: Fluent (IELTS Academic 7.0: Listening 7.0, Reading 7.0, Writing 7.0, Speaking 7.5) [\[eTRF\]](#)
Bengali: Native

| CERTIFICATIONS & COURSES

Certifications	ICCCAD SDG Policy Challenge; Millennium Fellowship 2024; Intl. Symposium on Marine Resource Management; Workshop on Small-Scale Fishers; Virtual Hackathon on Plastic-Free Rivers
Courses	Google (Crash Course on Python, Supervised ML); ICT Division (Intro to Python); 10 Minute School (Python for AI/ML); Reef Resilience (Adaptation Design, Remote Sensing); Bohubrihi (Python, Data Analytics); IUCN Red List (Modules 1–3)

| REFERENCES

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Thesis Supervisor

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Internship Supervisor